

Summary (max. 3,800 characters)

Transit systems are a key component of urban public policy. These systems intersect with broader transportation and urban land use issues and are therefore of great interest and practical import. In general terms, transit systems generate accessibility to opportunities that contribute to the economic operation of urban regions. But this is not the only service that transit systems provide, since they use space more efficiently than low-capacity vehicles (mainly private automobiles), and thus help to reduce congestion and result in lower emissions per capita. Furthermore, transit is an obligated form of transport for many members of the public who may not be able to own or use a private vehicle and increasingly supports many urban residents who choose not to own a vehicle. As such, transit systems also help to reduce transportation-related social exclusion, and support democratic participation in many essential activities.

Despite their importance in public policy, transit services are often offered without explicit mention of the principles that underpin their provision. Implicitly, transit is often seen as an economic service that must remain fiscally responsible. But while fiscal responsibility is an important consideration, it is not by any means the only one. In reality, many of the benefits of transit systems are non-monetary, and a tendency to see them mainly as economic services leads to a narrow approach that can blunt their attractiveness. Particularly, there are numerous examples where this leads to major inequities in terms of the distribution of the benefits and the costs of public transportation, while ignoring its role as a facilitator of democratic participation in society.

Planning for transit systems that are equitable and promote ideals of transportation equity and justice is an urgent task, especially as the limits of highly individualized transportation systems become clear. EquiTransit proposes to re-envision transit from a broader, more just perspective, and aims to create the knowledge and tools needed to realize this vision. This project follows on the heels of recent research initiatives in Canada, particularly the SSHRC-funded project Mobilizing Justice, which have broken new ground and made considerable progress towards more inclusive, equitable, and just transportation systems in Canada. EquiTransit builds on these foundations with a more targeted focus on public transportation.

Accordingly, the objective of EquiTransit is to generate novel, theoretically-sound and analytics-based equity indicators useful to assess the performance of public transit systems in Canada. Theoretically, we will use a framework of justice that emphasizes the 5WH approach (why, what, who, where, when, how). To ensure that our indicators are pertinent to the question of "why equity?", we propose a Codesign approach, whereby the research team will work with transit operators and users in the explicit identification of the rationale and motivation for the development and use of indicators. We plan to use cutting-edge geospatial technologies to investigate the questions "who seeks equity, and where and when?". Further, we propose to create open-source tools, including equity-informed planning and assessment tools, and place them at the fingertips of stakeholders at the lowest possible cost. Finally, in the spirit of Codesign, the effectiveness of the tools will be assessed in consultation with stakeholders, so that refinements can be incorporated in successive releases of the tools. In this way, EquiTransit aims to produce actionable knowledge to support the development of more equitable transit in Canada, to the benefit of urban residents, democratic governance, and democratic participation in Canadian society.

Detailed Description (max. 6 pages)

Objectives: Generate novel theoretically-sound and analytics-based equity indicators to assess the performance of public transit systems in Canada. Investigate their reliability by working with planners and transit users in areas with equity gaps. Create open-source tools, including dashboards, to make these indicators available to stakeholders. Assess the effectiveness and accessibility of the tools in consultation with stakeholders and incorporate feedback in newer releases of the tools.

Rationale: Traditional public transport performance evaluations usually rely on operational efficiency (1–3), reliability (4), safety (5), and objective comfort, but often leave aside equity considerations related to both the spatial and temporal coverage of transit service (6,7). Consequently, they often fail to meet the needs of the most disadvantaged residents (8). The transit planning process primary focus is on service cost (9–11), operational features (12–14), and emissions (12,15,16). Furthermore, current standards (e.g., Transit Capacity and Quality of Service Manual (TCQSM)) and manuals (e.g., Federal Transit Administration (FTA)) completely overlook equity dimensions, yet equity-deserving groups are those who rely most on public transit (17–20). When service is insufficient (21,22), they face the greatest risk of social isolation due to transport-related factors, preventing them from participating in essential activities (8,23,24). This group, characterized as captive-users, faces greater limitations caused by disparities in transit level of service (LoS), both spatially and temporally.

Approaches: The project aims to produce actionable knowledge to support the development of more equitable transit in Canada. At one level, we will produce equity indicators grounded in sound theories of justice. We consider this aspect of the project the one of greatest academic interest. Further, to ensure that the indicators are of practical relevance, they will be developed in partnership with stakeholders. A key aspect of procedural justice is ensuring that all stakeholders participate in the planning process and have access to the information they need for decision-making. For this reason, we: 1) adopt a codesign framework to design equity indicators in partnership with stakeholders in Hamilton, Ontario; and 2) propose to develop open-source tools to mobilize this knowledge. General Transit Feeds Specifications (GTFS) are the standard for evaluation and planning of transit operations, and will be used to implement an equity analytical module to complement the usual cost-based evaluation of transit systems. To facilitate the exploration of indicators and the evaluation and planning with the equity evaluation module, we will develop open-source tools (EquiTransit) that will embed all elements in an accessible interface (a dashboard). The tool will be based on open data and will be able to use data from different regions. The aim of the tools is to allow for the examination of socioeconomic and spatial-temporal inequalities in public transportation services, based on geo-spatial technologies.

Anticipated Benefits: The impacts of inequitable access to public transit are thought to be large, yet spatial-temporal inequalities in transit provision remain little understood and of peripheral interest in transit planning. Clear and transparent tools will allow these considerations to be integrated into planning practice and public discourse. Codesigning the indicators and tools can empower policy makers to make bolder decisions with greater public support. Including equity indicators in transit performance assessments will help uncover inequalities, enabling planners and advocates to understand how disadvantaged populations in different locations and time periods are, or are not, served by public transportation. For example, visualizations showing the frequency of public transit service across areas of different income levels and at different times of day can help practitioners identify where and when low-income populations are underserved, highlighting where interventions are most needed. Performance metrics will be disaggregated by income, race, gender, and other relevant population characteristics, in addition to their traditional spatial and temporal dimensions.

Facilitation of knowledge transition: This project aims to develop EquiTransit for planners and community members to better identify who has the most and least amounts of different transportation services, but even more crucially, where those populations are located and when are the services needed. Using spatial and temporal visualizations, planners and other stakeholders will be better able to address issues in equity in transportation access. This would be particularly beneficial for transit agencies to better understand where service changes or additions should be made to increase transit equity.

Public Transportation and Equity: Traditional transit performance evaluations optimize service for an average user by focusing on efficiency, ridership, and reliability. These are system level, operational performance metrics that frame transit evaluation in a way that does not engage with social outcomes. Yet, transit is most heavily relied upon by individuals who have the fewest alternatives (i.e., captive users). Evidence from Canadian sources supports this concern. In 2021, 17.6% of racialized commuters in large urban centres took public transit to work, compared with 6.4% of non-racialized commuters, and 13.1% of immigrants relied on transit versus 4.7% of those born in Canada (27). By May 2025, roughly one in four Black (25.7%), Latin American (25.3%), and Filipino (24.4%) commuters used transit, and the average transit commute had reached 44.1 minutes, nearly twice that of a car commute, with most racialized groups facing systematically longer journeys (28). These patterns coincide with material deprivation. Allen & Farber (25) estimate that close to one million Canadians across the eight largest CMAs live in low-income households in areas of poor transit accessibility, placing them at risk of transport poverty. Women in lower-income households also undertake substantially more care-related travel that are poorly serviced by current network designs (30).

This makes the distribution of service to different people, at diverse places, and various times a question of justice (25). Distributive justice in transportation asks who benefits from the system and who carries its burdens, shifting the evaluative criterion from system performance to the accessibility opportunities delivered to different socioeconomic and demographic groups (26). This framework moves beyond horizontal equity, which treats all riders identically, to vertical equity, that is, actively directing better service, coverage, and frequency to the groups who rely on it most and face the greatest systemic barriers. Vertical equity represents a stricter idea of justice, as new benefits are unequally distributed in favor of under-privileged populations.

Existing planning tools rarely delve into these inequities. A recent analysis of Toronto's bus network shows that although two-thirds of TTC riders are equity-deserving, most live outside the agency's designated equity priority areas, leaving much of the population the framework is meant to serve largely invisible to it (20). Traditional performance metrics, and even most accessibility measures, also average across the day, masking systematic temporal under-service during evenings, nights, and weekends, when shift workers, who are disproportionately racialized, immigrant, and low-income, most depend on transit (17). EquiTransit will close these gaps with indicators and tools that bring distributive justice into transit planning across both space and time.

WORK PACKAGES

WP.1 Theoretically-grounded equity indicators: It is well established that the burdens and benefits of transportation are distributed unevenly across space and time, with some groups receiving greater benefits at the expense of others (27). To better understand how these benefits and burdens are distributed among a diverse population, a theoretical framework related to equity and justice is needed. In this proposal, we follow recent work of the PI with members of the SSHRC-funded project Mobilizing Justice team (29) that adapted the framework of Jaggar (30) to the needs of transportation concerns. According to this framework, the examination of equity, fairness, and justice is guided by six interrelated questions: *Why?*, *Who?*, *What?*, *Where?*, *When?*, and *How?*, or the 5WH framework.

Why? Justice is a broad moral and political concept that relates to: (1) the rights and entitlements that should be recognised and enforced; (2) the fairness of processes and procedures in decision-making and distribution (procedural justice); and (3) the appropriate distribution of benefits and burdens of social life in society. The last concept, also known as distributive justice, is the most studied form of justice in the transportation literature and aims to provide a fair redistribution to repair past harms (31). This is not a technical question, but a question of values, and therefore it is incumbent on the research team to involve stakeholders to avoid imposing perspectives that may be extraneous to the community the research is meant to benefit. CODESIGN (WP.2) is essential to this purpose.

The questions of *Who?*, *What?*, *Where?*, and *When?* in the 5WH framework all intersect with the spatial and temporal distribution of transit and the populations it serves. For this reason, they are better investigated from a geo-spatial perspective (WP.3). These questions are described next.

Who? Disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, have become a key focus of transportation research (31,32). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences. Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (33–38).

What? In this research proposal, we argue that public transportation should be the focus of distributive justice. Public transportation is essential for promoting more sustainable cities due to its potential to reduce traffic congestion and carbon emissions and to improve air quality (39–41). In addition, public transportation increases people's accessibility, the main benefit of the transportation system (27), understood as the ease of reaching spatially distributed opportunities (42,43). In fact, public transportation provides access to opportunities for those who cannot - or prefer not to - drive, walk, or cycle. Access to opportunities and destinations is the most auspicious indicator for distributive justice (27), the most comprehensive indicator of the performance of land use and transport systems (44), and one of the main outcomes of spatial development (45) reduce the risk of social exclusion. Since transit and its output, accessibility, depend on space and time conditions, Scholars argue that this measure serves as a comprehensive and transparent indicator of transportation supply, with the additional advantage of being replicable across different jurisdictions and periods (45). Accessibility is a required condition for increasing people's freedom of choice and promoting equality of opportunities (45). An adequate level of access to opportunities, which can be offered by well-planned public transportation system, can increase the chances of attending a doctor's appointment, watching a movie at a cinema, engaging in a sporting activity in a park, or finding and keeping a job.

Where? By their very nature, interventions in the transportation sector are spatial interventions, and, due to the structuring effect of space (46), they inevitably generate inequality by benefitting those who live in or have access to the areas served by transportation infrastructure and services, while at the same time potentially harming others, either directly or indirectly (29). It is not uncommon for transportation policies to provide a sufficient or more than sufficient level of accessibility to the majority of people who own cars (47), while failing to ensure a basic level of accessibility for those who do not have access to a car, cannot use one, or do not wish to use one. To address this imbalance, improvements are needed in public transportation networks that provide users with a level of freedom comparable to that enjoyed by car users (48). The spatial dimension of analyses of transportation equity is often based on the concept of spatial equity, according to which cities should seek to distribute the benefits of transportation systems equally across space (31). Although transportation equity requires spatially targeted policies and interventions (34), rather than broad approaches that encompass all of society, our

perspective views equity primarily as an issue concerning people. Consequently, while we intend to identify areas with insufficient public transportation service and accessibility, these areas will be identified through measures and analyses focused on equity-deserving populations and facing insufficient access to public transportation, rather than solely on areas with low service. From this perspective, areas with a higher concentration of disadvantaged populations should take priority over wealthier areas, as disadvantaged groups tend to rely more on public transportation (28). Our work will be based on the City of Hamilton and its transit provider, Hamilton Street Railway (HSR).

When? To ensure justice in transportation, we evaluate the temporal aspects of transportation interventions. For instance, we investigate when a specific intervention takes place, the historical policy context in which it occurs, and for how long its benefits and burdens can be attributed to that intervention. In addition, the temporal dimension is central to our analysis because we are interested in investigating how equity-deserving populations may receive insufficient public transportation service, identifying not only the areas but also the times of day when this insufficiency is most pronounced.

The last question in our conceptual framework relates to How can measures be implemented to reduce inequities. The last work package of this project (WP.4) relates to the development of tools for equity-informed planning and evaluation. This part of the project intersects with WP.2 (CODESIGN) since these tools must implement concepts that are perceived as legitimate by all stakeholders.

How? Justice theories typically address sufficientarian and egalitarian moral concerns regarding how benefits and burdens should be distributed. While these two concepts are frequently applied in transportation equity research, they are often not explicitly articulated (22,49). The first approach, the sufficientarian perspective, focuses on why some groups or areas lack sufficient access to those goods and services. This approach builds on the idea of a ‘sufficient threshold’, that is, minimum levels of transport goods, services, and accessibility to essential activities that should be guaranteed to everyone to meet basic needs and lead to a meaningful life. In this idea, absolute priority should be given to improving the well-being of those whose well-being level falls below the sufficient threshold. Once that indicators have been selected and agreed upon by participants in the codesign part of the project, they will be implemented into equity-informed evaluation tools, as described in WP.4.

More generally, transportation analyses in practice often on standards to determine whether a situation is fair (29). Once operationalized, an equity standard defines the level of fairness to be adopted and the acceptable level of inequality, usually combining a concept of fairness with threshold criteria, thereby reflecting both vertical and horizontal approaches to equity. Vertical equity is typically operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute bus ride for older adults), whereas horizontal equity is generally operationalized through uniform travel barrier thresholds applied to all potential users (e.g., a maximum acceptable distance to a public transit stop). Regarding the definition of equity standards, our approach is guided by the principles of sufficiency and egalitarianism. This means that the minimum level of accessibility to public transportation should be defined according to the needs of the most equity-deserving population, thereby reducing inequalities in opportunities.

WP.2 Codesign: In the 5WH framework, the question of WHY? constitutes the cornerstone of our approach to analyzing equity. This question, however, is not answerable in a strictly technical fashion. Data can help to support it, but to produce truly actionable knowledge public transit users and practitioners must co-create the equity indicators developed to assess the performance of public transport systems in Canadian cities. For this reason, CODESIGN is an essential component of the project, and will run throughout the life-course of the project, from the development of indicators, the choice of standards, and the testing of open-source tools.

Codesign is a form of community-based participatory action research; it is a collaborative approach that brings together researchers and stakeholders to address the needs of the community, enable social system change, and increase the impact of research. Stakeholders are placed at the centre of the design process to effectively solve problems and improve human experiences (50–52). In this project, the stakeholders will include the research team, public transit providers, and users.

Though the tools for this overarching proposed research project will be developed for Canada's 14 largest CMAs, the Codesign process will focus on the City of Hamilton, where McMaster is located and where the strongest community-based research already partnerships exist. Research collaborators working at the HSR will be involved as the public transit providers. Public transit users will be selected from Hamilton's 'code red' neighbourhoods; areas that experience the highest rates of poverty, unemployment, and chronic illness in the city (53).

The research will follow the 5-step human centered Codesign approach (54). Empathy is the first step. We will begin by understanding service users' experiences with public transport systems as well as their understandings of the standards the currently exist. This will be done through a series of Codesign focus groups. The second and third steps, to Define the key issues, and to Ideate, i.e., to brainstorm possible solutions, will take place in a second round of focus groups. This time, a series of separate Codesign focus groups will take place with users and with transit providers. PhD student 1 will lead these three sets of focus groups and present the results to the larger research team. The larger team will then complete step 4, which is to Prototype a variety of tangible solutions based on the focus group discussions so far. The final step is to Test and re-design the solution as needed. A final round of focus groups with users and providers in which they will present the prototyped solutions for feedback. These results will be presented to the research team again, with the possibility of another round of re-design and testing. The final outputs should include new tools that are not only theoretically and empirically sound, but that also respond to the needs of transit users and providers.

All Codesign focus group sessions will be supported by McMaster university's equity-based Co creation hub (eqCC). This hub unites researchers, students, service providers, and users to engage educate and innovate for systemic transformation (55). All focus groups will take place at McMaster University, and all participants will be compensated for their time with a \$50 gift cards per focus group they participate in. They will also be reimbursed for transportation (either reimbursed a bus fare or parking will be paid for).

WP.3 Geospatial data analytics: EquiTransit will be built entirely from open data and open-source software, ensuring transparency and reproducibility across jurisdictions. For each Census Metropolitan Area, we assemble four open inputs: sociodemographic data from the 2026 Canadian Census (via the {cancensus} interface to CensusMapper from von Bergman et al. (56), General Transit Feed Specification (GTFS) schedules published by transit agencies, OpenStreetMap road and pedestrian networks, and the Global Human Settlement Layer built-up surface. To enable consistent comparison, these inputs are harmonised onto a fine hexagonal grid; dasymetric interpolation, restricted to residential cells, apportions census population and attributes to each hexagon. Using the {r5r} routing engine (57) we then compute different measures of transportation accessibility. This includes walking travel times between populated hexagons and transit stops to estimate the population effectively served by each stop. Larger network accessibility analyses will also be conducted to capture how well the transit system enables individuals to reach key destination types, such as places of work or daily necessities, around each study area within a reasonable travel time.

Critically, the travel time and accessibility analyses will be conducted using ten-minute transit service intervals, which is done to capture how level of service (e.g. service frequency) varies through the day and expose temporal gaps that daily averages conceal. GTFS Realtime data, which captures actual

transit vehicle movements rather than static schedules, offers a particularly interesting avenue to better understand how service levels and accessibilities are provided. Building on initial schedule-based analyses, we will expand our lens to incorporate realtime data over sampled time windows as the project progresses, making use of new computational tools like the {gtfsrealtime} (58) package for R. From this, the transit accessibilities are disaggregated by income, race, gender, and immigration status, revealing inequities across both space and time. The complete workflow will be packaged as a reproducible analytical pipeline and an interactive dashboard, initially covering Canada's fourteen largest CMAs, with parameters, indicators, and presentation refined through stakeholder consultation.

WP.4 Equity-informed transit planning and evaluation: Transit planning is often planned on legacy routes and updated as population and urban geography expand. Then the service is packaged into a global universal standard: General Transit Feed Specification (GTFS). While the planning process itself is mathematically challenging, it is fundamentally developed under a cost-reduction framework, maximizing the utilization of resources, and balancing operation with political decision and budget constraints. The well-established four-step transit design model (e.g., routes, frequencies, fleet, and crew assignment) does not include socio-economic and demographic factors that capture the equity dimension users. We will develop two interventions enabling a novel implementation of equity transit planning.

For existing network, a GTFS-based equity analytical model will be conceived, developed and deployed. The model will utilize equity indicator, geospatial data, and findings from the CODESIGN, and integrate them with GTFS files. Furthermore, Equity KPIs will be embedded in a GTFS-analytical model, enabling transit agencies, policymakers, and practitioners to evaluate the equity gaps in existing design. For new service design, the equity indicators will be integrated in the four-step transit model, offering a novel equity-based transit planning framework transferable to any context. Equity-indicators will be integrated in all four steps of the planning process. For example, frequency estimation is based on passenger flow/demand, wait-time, and fleet size (e.g., available resources). Although passenger flow/demand is an indirect indicator of equity, it remains ineffective of differentiating service provision for equity-deserving groups compared to other users.

Table T1. Project timeline with Work Packages and Tasks.

Work Packages	Tasks	Year 1	Year 2	Year 3	Year 4	Year 5
<i>WP.1 Theoretically-grounded equity indicators</i>	<i>Overall direction and transversal communication between work packages (Why? What? Who? Where? When? How?)</i>	Antonio Pérez & Juanita De Barros				
		PD 1				PD 2
<i>WP.2 Codesign</i>	<i>Equity indicators (Why?)</i>	Dr. Léa Ravensbergen				
	<i>EquiTransit tools (How?)</i>	PHD 1			PHD 1	PD 2
<i>WP.3 Geospatial data analytics</i>	<i>Equity indicators (What? Who? Where? When?)</i>	Dr. Christopher Higgins				
	<i>EquiTransit tools (How?)</i>	PD 1	PHD 2			PHD2
<i>WP.4 Equity-informed transit planning and evaluation</i>	<i>Development of base tools</i>	Dr. Moataz Mohamed				
	<i>Integration of equity indicators (How?)</i>	M 1			M 2	

Personnel (Key). PD1, 2: Post-Doctoral Fellows; PHD 1, 2: Doctoral Students; M1, 2: Master's Students

Knowledge Mobilization Plan (max. 1 page)

Appropriate Knowledge Mobilization (KM) strategies will be used to reach key audiences: 1) members of the public; 2) transportation planners and practitioners; and 3) academic and scientific peers. The creation of transparent, research-based tools will improve planning processes and outcomes, and support fluid participation in planning processes by a variety of stakeholders.

Public and practitioners (Years 1-5): A key aspect of the proposed work is the use of a codesign framework. In codesign there is a continuous and multilateral flow of knowledge during every step of the project. In addition to the involvement of members of the public and practitioners in this endeavor, we will expand the diffusion of this research by writing general reports in layperson language to facilitate knowledge transfer. These reports will take the form of policy briefs, user's manuals, and sample case reports related to the release of the tools. These will be posted publicly via a website that will be hosted on GitHub Pages. Members of the research team have experience developing and releasing available materials for public use with this medium (59,60). Communication with the public and practitioners will begin immediately through the codesign framework.

Academic (Years 2-5): This research proposal includes 4 research students (2 Master's and 2 PhDs), two post-doctoral researchers, five professors, and one international collaborator. The goal is for each Master's student to lead at least one paper for a national conference and then publish in an international peer-reviewed journal. PhD students will aim to publish three peer-reviewed journal articles. The post-doctoral fellows will lead one research paper each. Thus, 10 papers would be led by trainees. This project primarily has implications for planning in urban regions. Accordingly, target journals for this research include *Transportation Research A: Policy and Practice*; *Public Transportation*; *Transport Policy*; *Journal of Transport Geography*; and *Transportation*. Pre-print versions of papers will be published in their GitHub repositories. Conferences will target audiences with both researchers and practitioners. Further knowledge diffusion will be achieved via workshops in relevant conferences. Communication with academic audiences will begin in year 2 of the project when initial results can be reported.

Open-Science Statement: The Tri-Council system in Canada (i.e., CIHR, NSERC, SSHRC) is currently phasing in an Open Access policy for the research it funds. Open Access of publications and data are only one facet of open science, since encouraging quality of work throughout the research process requires an effort to share the data-collection process, methodology, and analysis as well (66). Open science can encourage and increase engagement from members of the public regarding ongoing research and any relevant issues that can emerge (61). In this respect, members of the research team have already a proven record of creating open, reproducible science, not only by publishing preprints publicly and sharing data, but by sharing every aspect of research, from the stage of data preparation and up to reproducible computational notebooks. Examples of data sharing by members of the research team include the creation of several open-data products (62) such as ActiveCA package (63) and TTS2016R package (59). An example of equity-based research in transportation that illustrates the mode of sharing pre-prints along with reproducible notebooks is Desjardins et al. (64). Materials such as this, shared publicly on websites like GitHub, allow for thorough verification of the research, to ensure it does not contain major errors that could impact the validity of the results (65). Open-science practices like this also speed up the process of discovery, by making it easy for other researchers to access the base work and modify it to accomplish whatever is required for their own specific work. The members of the research team are experienced open-science researchers and are fully committed to following Tri-Agency policy on Open Access, and more broadly to continue producing high-quality research that is open, transparent, and reproducible.

Research team, previous output and student training (max. 4 pages)

A. DESCRIPTION OF RESEARCH TEAM

The theme of public transportation and equity is multifaceted and requires an appropriate mix of skills and expertise. In response, this project gathers a group of researchers with complementary skills and perspectives. The research team is composed of one principal investigator, four co-investigators and one international collaborator, and includes scholars in the social sciences (Páez, Higgins, Ravensbergen, and Pereira), humanities (De Barros), and engineering (Páez and Mohamed). The principal investigator (Páez) and two of the co-investigators (Higgins and Ravensbergen) are alumni of the SSHRC-funded project Mobilizing Justice. One of the co-investigators (De Barros) is Director of McMaster's Centre for Human Rights and Restorative Justice, and another (Mohamed) is Director of McMaster Institute for Transportation and Logistics. Profiles and roles in the project are as follows.

Juanita DE BARROS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Professor of History at McMaster University and the director of the Centre for Human Rights and Restorative Justice. She is an internationally-recognized historian whose work explores the intersection of race, gender, and class in the construction of public policy, including in colonized urban spaces. This domain of knowledge is key to the project, as it combines social and government policy in ways that affect racialized and genderized people in urban spaces. De Barros expertise in discourse analysis and in the field of human rights is central to the aims of this project, where a clear understanding of stakeholders about the grounds for justice in public services is essential. For this project, she will lend her expertise and vision through the co-supervision of trainees.

Christopher D. HIGGINS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Associate Professor at the University of Toronto Scarborough in the Department of Human Geography and in the Department of Geography and Planning at the University of Toronto. He is among the world's top 2% of researchers in the Elsevier-Stanford rankings and received the 2025 University of Toronto Scarborough's New Scholar Award for exceptional research achievement and impact among early career faculty. He has experience working with public sector partners such as the City of Toronto, City of Hamilton, and Ontario Ministry of Transportation. His research uses geospatial analysis and accessibility measurement to examine how transportation systems shape access to housing, employment, and essential services, with particular attention to social equity. His work has been supported by SSHRC and NSERC, most recently through an NSERC Alliance Grant supporting the University of Toronto's tri-campus Mobility Network Consortium. For EquiTransit, he will lead the development of geospatial analytics, reproducible data pipelines, and interactive analytical tools.

Moataz MOHAMED (co-applicant, relative contribution to project 15%, 15% of personal research time) is associate Professor of Smart Systems and Transportation at McMaster University and Director of the McMaster Institute for Transportation and Logistics (MITL). He is the 2025 recipient of the Joseph Ip Distinguished Engineering Fellowship and is featured Elsevier-Stanford list of the world's top 2% of scientists. Mohamad is an experienced researcher, with more than 90 peer-reviewed articles and over \$6 million in secured research funding to his credit. He has trained 57 highly qualified personnel, including 11 PhD students, 12 master's students, and 7 postdoctoral fellows. Since 2018, he has been principal investigator on a City of Hamilton-funded program to reenvision the city's transit operator, Hamilton Street Railway (HSR). He is the principal developer of the AI-Transit Orchestrator, an \$1.4M NRCan-funded fleet-planning toolkit co-developed with Halifax Transit, Calgary Transit, Niagara Region Transit, and ETUQ. On this proposal, he will lead the team's work to integrate equity considerations in transit planning operations, translating multi-agency networks into a GTFS-based equity analytical model and equity-informed planning framework.

Antonio PÁEZ (principal applicant, relative contribution to project 30%, 80% of personal research time) is Professor of Transport Geography at McMaster University. He is a seasoned researcher who led the multi-institutional team that won the 2010 Meredith F. Burrill Award of the American Association of Geographers for exceptional work at the intersection of policy and basic research. He is also the recipient of the 2024 Edward L. Ullman Award of the American Association of Geographers for lifelong contributions to transportation geography. Páez has published widely, including over 170 peer-reviewed publications, three edited books, and a textbook. Páez is among the top 20 transportation researchers in the world whose careers started after 1999 according to Elsevier-Stanford's ranking, and is best known for his social science theory-informed quantitative research. This includes work on *normative accessibility* and *relative accessibility*, which aimed at implementing equity ideals in accessibility analysis. He is also well-versed in theoretical and qualitative approaches. He will direct the project and will participate in research and supervision of graduate students across the project.

Rafael H.M. PEREIRA (collaborator, relative contribution to project 10%, 5% of personal research time) is Director of Data Science at IPEA, Brazil's federal economics research institute. His contributions to transportation justice are largely responsible for shaping the field (27,32,66,67), while his contributions to numerous publicly available *R* and *Python* packages have been hugely influential in changing the practice of transportation research and planning. Open-source software such as {r5r} (57) and {r5py} (68) (which collectively have over a million downloads) are now routinely used by community planners, geographic researchers, and other members of the public for the purpose of simulating routing by different transportation modes. He has also worked on several packages for working with general transit feed specification data such as {gtfstools} (68). His role will be to provide advice on the theory and practice of transportation justice, and he will also lend his considerable expertise to the development of open-source tools.

Léa RAVENSBERGEN (co-applicant, relative contribution to project 15%, 15% of personal research time) is an early career researcher of equity in sustainable urban mobility. She is a mixed-methods researcher who has led multiple qualitative research projects focused on transportation users' experiences, including two that utilized a Codesign framework. In 2025, she led a series of Codesign focus groups with older adults on how best to design mobility reporting to encourage active lifestyles (currently under review in *Journal of Aging and Physical Activity*). In 2026, she worked with the Hamilton's transit operator Hamilton Street Railway (HSR) to Codesign how best to deliver paratransit as the City of Hamilton moves toward integrated transit. Ravensbergen will lead the Codesign component of the project.

B. DESCRIPTION OF PREVIOUS AND ONGOING RESULTS

Juanita DE BARROS's main field of research is into human rights and justice, with a focus on the political and social factors that shaped policies dealing with the intersection of health and the law in the past, their impact on the lived experience of women and children especially, and the longer-term impact of these policies. Her research projects include a sole-authored monograph on gender and reproductive health in the Caribbean (75), a forthcoming co-written article on urban vagrancy laws in late nineteenth-century Guyana, and a multidisciplinary co-edited essay collection (with McGill-Queens University Press). This latter project addresses child protection policies and discourse from the nineteenth to the early twenty-first centuries and the varied ways that these policies served adult institutional and political interests, often resulting in the forced labour and incarceration of children. This project is De Barros first foray into transportation-related research, and we believe that her historian's eye, in addition to her expertise in human rights and justice, will be an asset to the proposed work. Historical perspectives on

transportation remain rare despite their value (76,77) and collaboration between historians and transportation scholars is a unique aspect of the project.

Christopher D. HIGGINS has done research with conceptual, methodological, and applied outputs that are relevant to the proposed work. This includes the development of place-based/person-based (69), as well as competitive accessibility measures (43,70) that improve our understanding of who can reach destinations using the transportation network. His applied studies have incorporated equity into route-level transit performance measurement and examined disparities affecting the access of low-income residents, and essential workers to health and social services (38,71,72). Complementary research has assessed relationships among transit accessibility, residential location, and housing prices within individual metropolitan regions and across Canada (73). His ongoing research extends these approaches to the joint analysis of equity in accessibility and housing affordability. Further, Higgins brings a commitment to open, replicable, and reproducible research as evidenced through the development of the {pynogrid} package (74) and his work on a forthcoming open access book “Advanced Accessibility Analysis with R”. These skills mean that EquiTransit is well-grounded for developing theoretically-informed equity indicators, for conducting comparable analyses across Canadian metropolitan areas, and for communicating complex patterns through accessible analytical tools.

Moataz MOHAMED is an expert in developing and evaluating indicators of transit service performance and quality from both operator and rider perspectives (78–80). He has led qualitative and mixed-methods investigations into the needs of current and potential riders of transit, including grounded-theory studies of the barriers travelers face (81,82). As well, he has studied the satisfaction of transit users among population with distinct mobility needs (83). Mohamed is an expert in travel behaviour and preference heterogeneity across rider personas and generational cohorts (84–86), and in the development of accessibility and spatial-analysis measures for transportation (70,87,88). He has direct experience translating research into open, web-based decision-support tools for public agencies and practitioners, including work applying open-source data and machine learning to inform transportation decision-making (89). He has developed public-facing tools for the Ontario Ministry of Transportation. Currently, Mohamed is Director of the McMaster Institute for Transportation & Logistics and PI/Co-PI on a portfolio of NSERC-, NRCan-, and municipally funded projects on electric and zero-emission transit system planning, resilience, and equitable transit service design.

Antonio PÁEZ (Principal Investigator) has extensive experience with the study of the transportation challenges faced by transport-poor populations, including older adults (90–93), single parents (94,95), and children (96–98). His longstanding interest in the implications of social science theory and policy on methodology is illustrated by his research on normative accessibility (99) and relative accessibility (100), work initially motivated by the policy needs of Human Resources and Social Development Canada (an agency of Canada’s federal government). His interests now encompass broader issues of equity and justice in transportation (64,71,101). As a member of the SSHRC-funded project Mobilizing Justice, his research led to novel tools for the analysis of accessibility (43,70,102), clearer theoretical foundations for developing standards of equity (45), and the role of human dignity in transportation (103). Páez is an expert in the development and application of spatial analysis tools with a focus on transportation (87,94,104) and the development of web-based tools for public use (105,106), and he is also fluent in the use of qualitative methods (82,107). His breadth of knowledge and expertise is an asset for this project, in his role as PI and transversal conduit between the various work packages.

Léa RAVENSBERGEN has researched the transit experiences of equity-seeking groups (83,108,109) and the systemic barriers transit agencies face in planning just systems (108,110,111). Her expertise is as a qualitative methods researcher (109,112–114) including mobile ethnography (115,116). She has conducted Codesign research in which older adults codeveloped a mobility reporting tool (117) and has expertise in community-based research (118). Her work includes collaborations with all levels

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of governments in various provinces, transit agencies across the country, and non-for-profit organizations. She is part of a pan-Canadian team that was granted a \$.16 million NSERC Alliance grant to research data-driven tools for planning and operating gender-inclusive public transit systems in Canada. In this role, she is working in partnership with 14 Canadian transit agencies to better integrate and implement gender-sensitive data collection practices. Since 2023 she has worked with the Hamilton Street Railway (HSR), the public transit agency of the City of Hamilton (108,119,120). She is well positioned to lead to the Codesign element of this research, and has the relationships required to ensure a successful process.

C. DESCRIPTION OF TRAINING STRATEGIES

Our strategy for training highly qualified personnel is based on extensive involvement of the students, well-defined responsibilities for the integration of graduate students into all aspects of the project, and close supervision by a team of interdisciplinary supervisors (see Table MT.1).

Table MT.1. Supervisory Matrix

			Year	Main Supervisor	Supervisory Committee
WP.1 Theoretically-grounded equity indicators	Post-Doctoral Fellow 1	PD1	1	Páez	De Barros, Pereira
	Post-Doctoral Fellow 2	PD2	5	Páez	De Barros, Pereira
WP.2 Codesign	Ph.D. Student 1	PhD2	1 - 4	Ravensbergen	Páez, De Barros
WP.3 Geospatial data analytics	Ph.D. Student 2	PhD1	2 - 5	Higgins	Páez, Pereira
WP.4 Equity-informed transit planning and evaluation	Master Student 1	M1	1 - 2	Mohamed	Páez, Higgins
	Master Student 2	M2	3 - 4	Mohamed	Páez, Higgins

PD1 and PD2 will work with Páez and De Barros across work packages. PD1 will be part of WP.1 and will interact mainly with the geospatial data analytics package (WP.3) on the data analytics needed for the codesign work package (WP.2) to communicate with stakeholders. In the latter part of the project, PD2 will liaise primarily with the codesign package (WP.2) to assess the usability and performance (user experience) of EquiTransit tools. PhD and Master's students will maintain a tighter focus. PhD1 will be on the codesign package (WP.2) and PhD2 will focus on geospatial technologies and the development of open-source tools (WP.3). M1 and M2 will participate in the development of equity-based assessment and evaluation, beginning with a base tool that will incorporate equity-informed indicators after they have been validated as part of the codesign package (see **Table T1**). Through the project, all trainees will develop a variety of soft and transferable skills (see **Table MT.2**) through a combination of close supervision and regular discussion, and enrolment in appropriate graduate courses [e.g., reproducible research; (121)].

Table MT.2. Skills Matrix

	Year	KS	QL	QT	PP	DL	PC	DM	RE	CCE	TW	IIC	TM	BM
PD 1	1													
PD 2	5													
PhD 1	1 - 4													
M 1	1 - 2													
M 2	4 - 5													
PhD 2	2 - 5													

Skills (Key): KS: Knowledge Synthesis; QL: Qualitative research methods; QT: Quantitative research methods; PP: Publication and presentations (communication skills); DL: Digital literacy; PC: Programming and coding; DM: Data management and analysis; RE: research ethics; CCE: Consultation and community engagement; TW: Teamwork; IIC: International and intersectoral conference; TM: Time management; BM: Budget management.

Budget Justification (max. 2 pages)

The budget requested from SSHRC is \$499,540.

Personnel costs (\$464,000)

The largest portion of the costs are allocated towards the students (master's and PhD) and post-doctoral research fellows. Stipends for master's students are \$25,000 per year (years 1-2; 3-4), stipends of PhD students are \$28,000 per year (years 1-4; 2-5), and the Post-Doctoral Fellows will be paid \$70,000 per year (years 1 & 5). Over the duration of this 5-year project, this will account for 92.8% of the funds requested. A timeline of when master's students, PhD students, and post-doctoral fellows will be working on the project, and the tasks allocated to them, can be seen in **Table T1**.

Budget Item	Funds Requested
Student stipends	
Master's Student Stipends (4 student-years x \$25,000)	\$100,000
PhD Student Stipends (8 student-years x \$28,000)	\$224,000
Non-student salaries	
Post-Doctoral Salaries (2 person-years x \$70,000)	\$140,000
Total	\$464,000

Travel costs (\$21,000)

Beginning in year 2 of the project, each student will receive \$800 per year for Canadian travel to attend domestic conferences and meeting such as the Canadian Associations of Geographers Annual Conference or transportation-oriented conferences such as the Canadian Transportation Research Forum. This translates to \$2,400 in years 2-4 of the project, and \$800 in year 5, when the only student will be PhD2. Additionally, starting on year 2 of the project, each student will receive \$1,200 for international travel to conferences such as the Annual Meeting of the Transportation Research Board or NECTAR. This translates to \$3,900 per year in years 2 to 4 of the project and \$1,300 in year 5, when only PhD2 will be in the project. Regardless of year of study, students will be encouraged to attend the conferences as well as present in their upper years of study, as it is crucial for not only showcasing the research project, but also learning from other experts in the fields of transportation, geography, and planning. Hence, 4.2% of the total project cost will be allocated to travel costs.

Budget Item	Funds Requested
Travel and subsistence	
Students: Canadian travel (10 student-years x \$800)	\$8,000
Students: Foreign travel (10 student-years x \$1,300)	\$13,000
Total	\$21,000

Computer hardware (\$7,500)

The quantitative research segments of this project, namely the geospatial analysis, the development of the equity-informed evaluation tool, and the development of open-source packages, will require specialized equipment. This translates into computers with sufficient processors, RAM, and storage to be able to complete various computing processes and generating data. Members of the research team have some computers in their equipment inventory, and these will be used for this project. Funds are requested to upgrade or replace ageing equipment on years 2 and 3 of the project. We are budgeting \$3,500 in year 2 of the project, and \$3,000 in year 3 of the project to this end. In addition, we are

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requesting \$500 in each of years 4 and 5. These funds will be allocated to hardware updates, computer repairs, and the acquisition of various peripheral equipment as needed. Together with computer equipment already owned by members of the research team, the aim is to maintain four fully equipped workstations for the duration of the project, with one workstation assigned to each of the work packages.

Budget Item	Funds Requested
Non-disposable equipment	
Acquire computer equipment/replace ageing equipment (Year 2)	\$3,000
Acquire computer equipment/replace ageing equipment (Year 3)	\$3,000
Repair/peripherals (year 4)	\$500
Repair/peripherals (year 5)	\$500
Total	\$7,000

Other expenses: Codesign (\$7,540)

The qualitative research component of the project is the CODESIGN, with an estimated budget of \$6,000 to provide incentives for participants across different focus groups (based on factors such as socioeconomic status and/or neighbourhood). Each codesign focus group participant will receive a \$50 gift card and their transport to and from McMaster reimbursed ($\$4.00 \times 2$ (return bus fare) = \$8.00). All co-design focus group materials (e.g., printing, Bristol boards, etc.) will be provided by Ravensbergen's lab in kind. There will be two codesign phases: development of indicators and development of tools.

Budget Item	Funds Requested
Other: Codesign (Development of indicators)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 2) ($4 \times 5 \times \$58$)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 2) ($3 \times 5 \times \$58 + 2 \times 5 \times \58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 3)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 3) ($3 \times 4 \times \$58 + 2 \times 4 \times \58)	\$1,160
Other: Codesign (Development of tools)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 4) ($4 \times 5 \times \$58$)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 4) ($3 \times 5 \times \$58 + 2 \times 5 \times \58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 5)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 5) ($3 \times 4 \times \$58 + 2 \times 4 \times \58)	\$1,160
Total	\$7,540

Software (\$0)

This project will exclusively use open-source software.

Open access publishing (\$0)

We will leverage publishing agreements made by our institutions with reputable publishers such as Wiley, PLoS ONE, Elsevier, and Springer.

Expected outcomes (max. 3,800 characters)

The project assembles a group of researchers with a proven record of collaboration, and it fosters new collaborations between several research professors. This includes new collaborative opportunities between Ravensbergen and Higgins, Mohamed and Pereira, Ravensbergen and Pereira, and all transportation scholars collaborating with De Barros, an expert in human rights and justice. The team takes previous collaborations and expands them in novel ways that will benefit the quality of the scholarship produced. Moreover, the codesign framework proposed builds on existing collaborations (e.g., between Co-PI Ravensbergen and Mohammed and Hamilton's transit agency) and fosters new connections with community stakeholders. The project aspires to empower users of transit to have a more fluid participation in decision-making processes that affect them, and to help providers of a key public service to better incorporate user knowledge. In this way, the codesign framework opens avenues for knowledge exchange between providers and users of transit and thus contributes to enhance the process and practice of effective governance.

Through this project, four graduate students and two post-doctoral researchers will be trained and gain various new skills, to the betterment of Canada's transportation and research expertise. Other outcomes of the project are of three kinds: communication, research materials, and tools. The project will produce ten peer-reviewed papers (1 article by each Masters student, 3 articles per Doctoral student, and 1 article by each post-doctoral researcher). In addition, there will be communications for lay audiences, in the form of case studies, policy briefs, and user's manuals. New data produced during the research will be shared in the form of open data products, which will be made available via web repositories in addition to registering them for digital object identification (doi) using Zenodo. Other research materials, such as research documents and computational notebooks, will be shared using public web repositories. The project aims to push the boundaries of current practices in open science by expanding on the current expertise of members of the research team. The project will also produce open-source tools (e.g., dashboards) to help stakeholders assess the equity implications of various transit provision scenarios. More generally, knowledge transfer will prepare the next generation of transportation equity champions.

With better knowledge, better choices will be facilitated for planners and policymakers at transit agencies and users of public transportation. Rather than basing decisions on relatively narrow fiscal considerations, transparent and publicly available tools will contribute to advance more equitable public transportation planning. As a matter of procedural justice in transportation, planning must consider all segments of society, and for participation to be meaningful relevant information must be accessible. The tools proposed contribute to fill an information gap about levels of service by clarifying in a more holistic way the distribution of transit's costs and benefits. The development of visual tools based on research that accounts for relevant perspectives of stakeholders will enhance public discourse, public policy, and planning. Generating knowledge about the equity implications of transit operations will help steer conversations from a predominantly economic perspective to a fuller accounting of transit's numerous non-monetary benefits. Generating the tools to implement this knowledge will help inform policy and practice related to urban planning, transport, social inclusion, and community development by translating abstract research into trustworthy, locally relevant, and actionable information.

Research Contributions: Antonio Páez

Refereed contributions BOOKS 2020-2026 (*: SSHRC supported; +: Student author)

1. Páez, A. & Boisjoly, G. (2026). *Discrete Choice Analysis* with R. Springer Cham.
<https://doi.org/10.1007/978-3-031-20719-8>

Refereed contributions JOURNAL ARTICLES 2020-2026 (*: SSHRC supported; +: Student author) (54 peer-reviewed journal publications in this period, only 30 most relevant to proposal are listed below)

- * 1. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2026). A 30-year analysis of active travel behaviour in Canada. *Journal of Transport Geography*, 131, 104561.
<https://doi.org/10.1016/j.jtrangeo.2026.104561>
- * 2. Higgins, C. D., Li, C., & Páez, A. (2026). Person-based aggregate space-time accessibility (PASTA): Bridging the gap between place- and person-based accessibility. *Journal of Transport Geography*, 132, 104602.
<https://doi.org/10.1016/j.jtrangeo.2026.104602>
- * 3. Soukhov, A. (+) Pereira, R. H. M., Higgins, C. D., & Páez, A. (2025). A family of accessibility measures derived from spatial interaction principles. *PLOS ONE*, 20(11), 1–31. <https://doi.org/10.1371/journal.pone.0335951>
- * 4. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2025). ActiveCA: Time use data from the general social survey of Canada to study active travel. *Environment and Planning B: Urban Analytics and City Science*, 52(8), 2037–2047.
<https://doi.org/10.1177/23998083251374724>
- * 5. Soukhov, A., Ravensbergen, L., Dorantes, L. M., & Páez, A. (2025). Towards completely caring 15-minute neighbourhoods. *Transportation Research Part A: Policy and Practice*, 198, 104503. <https://doi.org/10.1016/j.tra.2025.104503>
6. Elizondo-Candanedo, R. F. (+), Arranz-López, A., Soria-Lara, J. A., & Páez, A. (2024). When e-activities meet spatial accessibility: A theoretical framework and empirical space-time thresholds for simulated spatial settings. *Journal of Transport Geography*, 121, 104026. <https://doi.org/10.1016/j.jtrangeo.2024.104026>
- * 7. Tavakoli, Z., Waygood, O., Abdollahi, S., & Páez, A. (2024). “Where Do Children Go?”: Exploring Children’s Daily Destinations with Children, Parents, and Experts. *Urban Planning*, 9(0). <https://doi.org/10.17645/up.8478>
- * 8. Tavakoli, Z. (+), Abdollahi, S. (+), Waygood, E. O. D., Páez, A., & Boisjoly, G. (2024). Traffic danger’s potential impact on children’s accessibility. *Transportation Research Part D: Transport and Environment*, 135, 104370. <https://doi.org/10.1016/j.trd.2024.104370>
9. Jamal, S. (+), & Páez, A. (2024). Well-being implications of immobility during COVID-19: Evidence from a student sample in Bangladesh using the satisfaction with life scale. *Transportation*, 51(5), 2019–2049. <https://doi.org/10.1007/s11116-023-10395-z>
- * 10. Desjardins, E., Lam, J., Reynard, D., Collins, D., Waygood, E.O.D., & Páez, A. (2024). Framing active school travel in Ontario, or how spinach is good for you. *Transportation Research Part A: Policy and Practice*, 180, 103953.
<https://doi.org/10.1016/j.tra.2024.103953>
- * 11. Soukhov, A. (+), Tarriño-Ortiz, J. (+), Soria-Lara, J. A., & Páez, A. (2024). Multimodal spatial availability: A singly-constrained measure of accessibility considering multiple modes. *PLOS ONE*, 19(2), e0299077. <https://doi.org/10.1371/journal.pone.0299077>
12. Rey-Blanco, D., Arbués, P., López, F. A., & Páez, A. (2024). Using machine learning to identify spatial market segments. A reproducible study of major Spanish markets. *Environment and Planning B: Urban Analytics and City Science*, 51(1), 89–108.
<https://doi.org/10.1177/23998083231166952>

13. Merrall, J. (+), Higgins, C. D., & **Páez, A.** (2024). What's a school worth to a neighborhood? A spatial hedonic analysis of property prices in the context of accommodation reviews in Ontario. *Geographical Analysis*, 56(2), 217–243.
* <https://doi.org/https://doi.org/10.1111/gean.12377>
14. Soukhov, A. (+), **Páez, A.**, Higgins, C. D., & Mohamed, M. (2023). Introducing spatial availability, a singly-constrained measure of competitive accessibility. *PLOS ONE*, 18(1), e0278468. <https://doi.org/10.1371/journal.pone.0278468>
15. Soukhov, A. (+), & **Páez, A.** (2023). TTS2016R: A data set to study population and employment patterns from the 2016 Transportation Tomorrow Survey in the Greater Golden Horseshoe Area, Ontario, Canada. *Environment and Planning B: Urban Analytics and City Science*, 50(2), 556–563. <https://doi.org/10.1177/23998083221146781>
16. Sears, S. (+), Moataz, M., Ferguson, M., Razavi, S., & **Páez, A.** (2022). Perceived barriers to the movement of goods in Canada: A grounded theory investigation. *Transportation Research Part A: Policy and Practice*, 162, 27–45.
* <https://doi.org/10.1016/j.tra.2022.05.011>
17. Desjardins, E. (+), Tavakoli, Z. (+), **Páez, A.**, & Waygood, E. O. (2022). Children's access to non-school destinations by active or independent travel: A scoping review. In *International Journal of Environmental Research and Public Health*, No. 19; Vol. 19. <https://doi.org/10.3390/ijerph191912345>
18. Desjardins, E. (+), Higgins, C. D., Scott, D. M., Apatu, E., & **Páez, A.** (2022). Correlates of bicycling trip flows in Hamilton, Ontario: Fastest, quietest, or balanced routes? *Transportation*, 49(3), 867–895. <https://doi.org/10.1007/s11116-021-10197-1>
19. Desjardins, E. (+), Higgins, C. D., & **Páez, A.** (2022). Examining equity in accessibility to bike share: A balanced floating catchment area approach. *Transportation Research Part D: Transport and Environment*, 102, 103091. <https://doi.org/https://doi.org/10.1016/j.trd.2021.103091>
20. Demitiry, M. (+), Higgins, C. D., **Páez, A.**, & Miller, E. J. (2022). Accessibility to primary care physicians: Comparing floating catchments with a utility-based approach. *Journal of Transport Geography*, 101, 103356. <https://doi.org/10.1016/j.jtrangeo.2022.103356>
21. **Páez, A.**, & Higgins, C. D. (2021). An examination of the accessibility implications of a pilot COVID-19 vaccination program in Hamilton, Ontario. Findings. <https://doi.org/10.32866/001c.24082>
22. **Páez, A.** (2021). Open spatial sciences: An introduction. *Journal of Geographical Systems*, 23(4), 467–476. <https://doi.org/10.1007/s10109-021-00364-4>
23. Pereira, R. H. M., Braga, C. K. V., Servo, L. M., Serra, B., Amaral, P., Gouveia, N., & **Páez, A.** (2021). Geographic access to COVID-19 healthcare in Brazil using a balanced float catchment area approach. *Social Science & Medicine*, 273. <https://doi.org/10.1016/j.socscimed.2021.113773>
24. Higgins, C. D., **Páez, A.**, Kim, G. (+), & Wang, J. (2021). Changes in accessibility to emergency and community food services during COVID-19 and implications for low income populations in Hamilton, Ontario. *Social Science & Medicine*, 291. <https://doi.org/10.1016/j.socscimed.2021.114442>
25. Eldeeb, G. (+), Mohamed, M., & **Páez, A.** (2021). Built for active travel? Investigating the contextual effects of the built environment on transportation mode choice. *Journal of Transport Geography*, 96. <https://doi.org/10.1016/j.jtrangeo.2021.103158>
26. Desjardins, E. (+), Higgins, C. D., Scott, D. M., Apatu, E., & **Páez, A.** (2021). Using environmental audits and photo-journeys to compare objective attributes and bicyclists' perceptions of bicycle routes. *Journal of Transport & Health*, 22. <https://doi.org/10.1016/j.jth.2021.101092>

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27. Desjardins, E. (+), Apatu, E., Razavi, S. D., Higgins, C. D., Scott, D. M., & **Páez, A.** (2021). “Going through a little bit of growing pains”: A qualitative study of the factors that influence the route choice of regular bicyclists in a developing cycling city. *Transportation Research Part F-Traffic Psychology and Behaviour*, 81, 431–444. <https://doi.org/10.1016/j.trf.2021.06.005>
28. Sarlas, G. (+), **Páez, A.**, & Axhausen, K. W. (2020). Betweenness-accessibility: Estimating impacts of accessibility on networks. *Journal of Transport Geography*, 84. <https://doi.org/10.1016/j.jtrangeo.2020.102680>
29. Jamal, S. (+), & **Páez, A.** (2020). Changes in trip-making frequency by mode during the COVID-19 emergency in Bangladesh. *Findings*. <https://doi.org/10.32866/001c.17977>
30. Jamal, S. (+), Mohiuddin, H., & **Páez, A.** (2020). How do the perceptions of neighborhood conditions impact active transportation? A study in Rajshahi, Bangladesh. *Transportation Research Part D-Transport and Environment*, 87. <https://doi.org/10.1016/j.trd.2020.102525>

Forthcoming contributions JOURNAL ARTICLES (*: SSHRC supported; +: Student author)

1. Soukhov, A. (+), Tiznado-Aitken, I., Palm, M., Farber, S., & **Paez, A.** (2025). Standards of fairness in the transportation literature. *Transportation*. <https://doi.org/10.1007/s11116-025-10693-8> (in press)
2. Soukhov, A. (+), Higgins, C. D., Mohamed, M., & **Páez, A.** (2025). Ten years of school closures and consolidations in Hamilton, Canada and the impact on multimodal accessibility. *Spatial Networks and Economics*. <https://doi.org/10.1007/s11067-025-09677-z> (in press)
3. Dos Santos, B.D. (+), Hansen, T., & **Páez, A.** (2026). Transportation and dignity: A scoping review and theoretical framework. *Transport Reviews* (revised and submitted)
4. Jamal, S., & **Páez, A.** (2026). Changes in walking frequency and its equity implications during the first wave of COVID-19 in Bangladesh. *Journal of Transport & Health* (revised and submitted)
5. Dos Santos, B.D. (+), Soukhov, A., & **Páez, A.** (2026) Equity Standards for transport policy: using objective accessibility, perceived accessibility and employment outcomes to set thresholds for equity planning. *Transportation Research Part A: Policy and Practice* (submitted)

Other research contributions (2020-2026)

- Between 2020-2026 I co-authored two reports (for Mobilizing Justice) and two book chapters.
- Editor-in-Chief, *Journal of Geographical Systems*.
- Editorial Board Member (*Annals of the Association of American Geographers*; *Economic and Business Letters*; *Geographical Analysis*; *International Journal of Geographical Information Science*; *Papers in Regional Science*; *Transportation*).
- 36 invited talks (17: Canada, 6: Spain, 4: Brazil, 3: USA, 2: Australia, 2: UK, 1: Italy, 1: Portugal).

Most significant career research contributions

A total of 177 publications in my record are indexed in Scopus. These publications have been cited 8,473 times (as of September 23, 2026). The average number of citations per indexed publication is 47.87, and the h-index is 50 (50 publications have been cited 50 times or more). Most significant career contributions are:

1. **Páez A**, Scott DM, Morency C (2012). Measuring accessibility: positive and normative implementations of various accessibility indicators, *Journal of Transport Geography*, 25 141-153. 141–153. <https://doi.org/10.1016/j.jtrangeo.2012.03.016> (**656 citations**). This research was key to

distinguishing between behavioral processes and policy aims in accessibility analysis. A key finding was the identification of critical issues in the way accessibility measures are interpreted depending on their implementation. By providing concrete guidance for interpretation, this paper has become a key reference in accessibility research and has been cited in the fields of transportation, geography, economics, environmental studies, civil engineering, urban studies, and operations research.

2. **Páez, A., & Whalen, K. (+)** (2010). Enjoyment of commute: A comparison of different transportation modes. *Transportation Research Part A-Policy and Practice*, 44(7), 537–549. <https://doi.org/10.1016/j.tra.2010.04.003> (**273 citations**). This paper provided evidence of the intrinsic utility of active travel and challenged the conventional wisdom that travel time is always a disutility, with important implications for policy. Kate Whale was a Master student when this paper was written. It has become a valued reference for research in the fields of transportation, geography, business economics, public administration, and engineering.
3. **Páez A, Mercado RG (+), Farber S. (+), Morency C, Roorda M** (2010). Relative Accessibility Deprivation Indicators for Urban Settings: Definitions and Application to Food Deserts in Montreal, *Urban Studies*, 47 (7) 1415-1438. <https://doi.org/10.1177/0042098009353626> (**234 citations**). The methods presented in this paper were an early attempt at obtaining benchmarks for equity analysis. They provided the tools to assess differences in accessibility from a selected baseline, similar to the way poverty lines are measured with respect to some judiciously selected “median household”. This paper has become an important resource for research in geography, transportation, economics, environmental studies, public health, and urban studies.
4. Reyes, M.(+), **Páez, A., & Morency, C.** (2014). Walking accessibility to urban parks by children: A case study of Montreal. *Landscape and Urban Planning*, 125, 38–47. <https://doi.org/10.1016/j.landurbplan.2014.02.002> (**196 citations**). This research was led by Mario Reyes as a Master student, and it was among the first papers to examine the provision of green spaces from the perspective of children. This paper has been cited in the fields of environmental sciences, economics, public health, geography, and public administration.
5. Black J, **Páez A, Suthanaya P, (+)** (2002), Sustainable Urban Transportation: Performance Indicators and Some Analytical Approaches, *ASCE Journal of Urban Planning and Development*, 128 (4) 184-209 (**195 citations**). Sustainability has long been an interest of mine, particularly the social pillar of sustainability. This collaboration with John Black and Putu Suthanaya has been influential in civil engineering, transportation, urban studies, economics, and environmental science.

Contributions to training

Since 2020, I have supervised 8 senior undergraduate students in the programs of Engineering and Society (3), Geography (2), Life Sciences (2), and Economics (1). All these students completed a capstone project or thesis. I supervised 4 master’s students (3 in Geography, 1 in Public Health) and cosupervised 3 master students (2 in Geography and 1 in eHealth). I supervised 2 doctoral students (both in Geography, including a recipient of a Canada Graduate Scholarship award) and cosupervised 1 doctoral student in Civil Engineering. I currently supervise two master’s students in Geography and two doctoral students in Geography (including 1 Vanier Scholar). One master student (Desjardins) was the winner of the Master’s Thesis Competition in Transport Geography by the American Association of Geographers (2021). One doctoral student (Soukhov) was the winner of the Doctoral Thesis Competition in Transport Geography by the Association of American Geographers (2025). Graduate students working with me always are offered opportunities to participate in research projects. Graduate students have participated in research leading to 37 out of a total of 54 peer-reviewed publications on record in the period since 2020, and their contributions are recognized with coauthorships.

Special circumstances

Since starting at McMaster in 2002, I have had four episodes of major depression. The first three were in 2005, 2013, and 2015. Unwisely, instead of taking the time necessary to heal at any of those opportunities, I pushed ahead with a work schedule that distracted me, but did not address my health needs. In 2025 I had another episode of major depression that was darker and longer than any of the preceding ones. Finally, I realized that I needed to recover, and was on medical leave from the university from November, 2025, until April 2026. This period of time away from work accounts for a gap in my teaching record and a decrease in my research activities, including grant writing and publishing.

List of References (max. 10 pages)

1. Mazloumi E, Currie G, Rose G. Using GPS Data to Gain Insight into Public Transport Travel Time Variability. *J Transp Eng.* 2010 Jul;136(7):623–31. doi:10.1061/(ASCE)TE.1943-5436.0000126
2. Sami J, Pascal F, Younes B. Public Road Transport Efficiency: A Stochastic Frontier Analysis. *Journal of Transportation Systems Engineering and Information Technology.* 2013 Oct 1;13(5):64–71. doi:10.1016/S1570-6672(13)60123-3
3. Zhang L, Liu Y, Li S, Chen H, Liu Y. Evaluating public transport performance to guide public transport planning: A unified prior method. *Travel Behaviour and Society.* 2023 Oct 1;33:100605. doi:10.1016/j.tbs.2023.100605
4. Trompet M, Liu X, Graham DJ. Development of Key Performance Indicator to Compare Regularity of Service between Urban Bus Operators. *Transportation Research Record: Journal of the Transportation Research Board.* 2011 Jan;2216(1):33–41. doi:10.3141/2216-04
5. Joewono TB, Kubota H. SAFETY AND SECURITY IMPROVEMENT IN PUBLIC TRANSPORTATION BASED ON PUBLIC PERCEPTION IN DEVELOPING COUNTRIES. *IATSS Research.* 2006 Jan 1;30(1):86–100. doi:10.1016/S0386-1112(14)60159-X
6. Adli SN, Chowdhury S. A Critical Review of Social Justice Theories in Public Transit Planning. *Sustainability.* 2021 Jan;13(8):4289. doi:10.3390/su13084289
7. İmre Ş, Çelebi D. Measuring Comfort in Public Transport: A case study for İstanbul. *Transportation Research Procedia.* 2017 Jan 1;World Conference on Transport Research - WCTR 2016 Shanghai. 10-15 July 201625:2441–9. doi:10.1016/j.trpro.2017.05.261
8. Lucas K. Transport and social exclusion: Where are we now? *Transport Policy.* 2012 Mar 1;URBAN TRANSPORT INITIATIVES20:105–13. doi:10.1016/j.tranpol.2012.01.013
9. Calabrò G, Pira ML, Giuffrida N, Inturri G, Ignaccolo M, Correia GH de A. Designing demand responsive transport services in small-sized cities using an agent-based model. *Transportation Research Procedia.* 2023 Jan 1;AIIT 3rd International Conference on Transport Infrastructure and Systems (TIS ROMA 2022), 15th-16th September 2022, Rome, Italy69:759–66. doi:10.1016/j.trpro.2023.02.233
10. Currie G, Fournier N. Why most DRT/Micro-Transits fail – What the survivors tell us about progress. *Research in Transportation Economics.* 2020 Nov 1;Thredbo 16 conference83:100895. doi:10.1016/j.retrec.2020.100895
11. Mishra S, Mehran B, Sahu PK. Assessment of delivery models for semi-flexible transit operation in low-demand conditions. *Transport Policy.* 2020 Dec 1;99:275–87. doi:10.1016/j.tranpol.2020.09.004
12. Bürstlein J, López D, Farooq B. Exploring first-mile on-demand transit solutions for North American suburbia: A case study of Markham, Canada. *Transportation Research Part A: Policy and Practice.* 2021 Nov 1;153:261–83. doi:10.1016/j.tra.2021.08.018
13. Davison L, Enoch M, Ryley T, Quddus M, Wang C. A survey of Demand Responsive Transport in Great Britain. *Transport Policy.* 2014 Jan 1;31:47–54. doi:10.1016/j.tranpol.2013.11.004
14. Nelson JD, Wright S, Masson B, Ambrosino G, Naniopoulos A. Recent developments in Flexible Transport Services. *Research in Transportation Economics.* 2010 Jan 1;Reforming Public Transport throughout the World29(1):243–8. doi:10.1016/j.retrec.2010.07.030
15. Diana M, Quadrifoglio L, Pronello C. Emissions of demand responsive services as an alternative to conventional transit systems. *Transportation Research Part D: Transport and Environment.* 2007 May 1;12(3):183–8. doi:10.1016/j.trd.2007.01.009

16. Dytckov S, Persson JA, Lorig F, Davidsson P. Potential Benefits of Demand Responsive Transport in Rural Areas: A Simulation Study in Lolland, Denmark. *Sustainability*. 2022 Jan;14(6):3252. doi:10.3390/su14063252
17. Palm M, Allen J, Farber S. Shifted out: The well-being and justice implications of evening and night commuting. *Transportation research Part D, Transport and environment*. 2023;122:103875-. doi:10.1016/j.trd.2023.103875
18. Ravensbergen L, Fournier J, El-Geneidy A. Exploratory Analysis of Mobility of Care in Montreal, Canada. *Transportation Research Record: Journal of the Transportation Research Board*. 2023 Jan;2677(1):1499–509. doi:10.1177/03611981221105070
19. Statistics Canada G of C. The Daily — Number of Canadian commuters increases for fourth straight year in 2025 [Internet]. 2025 [cited 2026 May 21]. Available from: <https://www150.statcan.gc.ca/n1/daily-quotidien/250826/dq250826a-eng.htm>
20. Tabascio A, Tiznado-Aitken I, Higgins C, Farber S. Incorporating equity into transit performance measures: A disaggregated bus route level approach. *Case studies on transport policy*. 2024;17:101256-. doi:10.1016/j.cstp.2024.101256
21. Luz G, Portugal L da S. Understanding transport-related social exclusion through the lens of capabilities approach. *Transport Reviews*. 2022 Jul 1;42(4):503–25. doi:10.1080/01441647.2021.2005183
22. Martens K. Defining sufficient accessibility. In: *Transport Justice*. Routledge; 2016.
23. Statistics Canada G of C. The Daily — Has the COVID-19 pandemic changed commuting patterns for good? [Internet]. 2022 [cited 2026 May 21]. Available from: <https://www150.statcan.gc.ca/n1/daily-quotidien/221130/dq221130c-eng.htm>
24. Statistics Canada G of C. The Daily — Number of Canadian commuters increases for fourth straight year in 2025 [Internet]. 2025 [cited 2026 May 21]. Available from: <https://www150.statcan.gc.ca/n1/daily-quotidien/250826/dq250826a-eng.htm>
25. Allen J, Farber S. Sizing up transport poverty: A national scale accounting of low-income households suffering from inaccessibility in Canada, and what to do about it. *Transport policy*. 2019;74:214–23. doi:10.1016/j.tranpol.2018.11.018
26. Ravensbergen L, Fournier J, El-Geneidy A. Exploratory Analysis of Mobility of Care in Montreal, Canada. *Transportation research record*. 2023;2677(1):1499–509. doi:10.1177/03611981221105070
27. Pereira RHM, Schwanen T, Banister D. Distributive justice and equity in transportation. *Transport Reviews*. 2017;37(2):170–91. doi:10.1080/01441647.2016.1257660
28. Karner A, Pereira RHM, Farber S. Advances and pitfalls in measuring transportation equity. *Transportation (Dordrecht)*. 2025;52(4):1399–427. doi:10.1007/s11116-023-10460-7
29. Soukhov, Anastasia, Tiznado-Aitken, Ignacio, Palm, Matthew, Farber, Steven, Páez, Antonio. Searching for standards of fairness in the transportation justice literature [Internet]. *Mobilizing Justice*; 2024 Jan [cited 2024 Sep 27]. p. 22. Report MJ-0001. Available from: <https://mobilizingjustice.ca/wp-content/uploads/2024/01/tools-report-2.pdf>
30. Jaggar AM. The philosophical challenges of global gender justice. *Philosophical Topics*. 2009;37(2):1+. Located at: Gale Academic OneFile
31. Lucas K. A new evolution for transport-related social exclusion research? *Journal of Transport Geography*. 2019 Dec 1;Celebrating 25 years of Journal of Transport Geography81:102529. doi:10.1016/j.jtrangeo.2019.102529

32. Pereira RHM, Karner A. Transportation Equity. In: Vickerman R, editor. *International Encyclopedia of Transportation* [Internet]. Oxford: Elsevier; 2021 [cited 2024 Sep 24]. p. 271–7. Available from: <https://www.sciencedirect.com/science/article/pii/B9780081026717100533> doi:10.1016/B978-0-08-102671-7.10053-3
33. Bania N, Leete L, Coulton C. Job Access, Employment and Earnings: Outcomes for Welfare Leavers in a US Urban Labour Market. *Urban Studies*. 2008 Oct 1;45(11):2179–202. doi:10.1177/0042098008095864
34. Martens K, Bastiaanssen J, Lucas K. 2 - Measuring transport equity: Key components, framings and metrics. In: Lucas K, Martens K, Di Ciommo F, Dupont-Kieffer A, editors. *Measuring Transport Equity* [Internet]. Elsevier; 2019 [cited 2024 Nov 1]. p. 13–36. Available from: <https://www.sciencedirect.com/science/article/pii/B9780128148181000020> doi:10.1016/B978-0-12-814818-1.00002-0
35. Merlin LA, Hu L. Does competition matter in measures of job accessibility? Explaining employment in Los Angeles. *Journal of Transport Geography*. 2017 Oct 1;64:77–88. doi:10.1016/j.jtrangeo.2017.08.009
36. Middleton J, Spinney J. 4 Social inclusion, accessibility and emotional work. In: Docherty I, Shaw J, editors. *Transport Matters* [Internet]. Policy Press; 2019 [cited 2025 Feb 13]. p. 83–104. Available from: <https://www.degruyter.com/document/doi/10.56687/9781447329572-009/html?srsIid=AfmBOoqjdOQp3sJesUxgtJPEfg6LVN1eBYG9y6E9ev22CTBXZAruyRCd> doi:10.56687/9781447329572-009
37. Soukhov A, Mooney N, Ravensbergen L. Exploring mobility of care with measures of accessibility. *Journal of Transport Geography*. 2025 Feb 1;123:104050. doi:10.1016/j.jtrangeo.2024.104050
38. Tabascio A, Tiznado-Aitken I, Higgins C, Farber S. Incorporating equity into transit performance measures: A disaggregated bus route level approach. *Case Studies on Transport Policy*. 2024 Sep 1;17:101256. doi:10.1016/j.cstp.2024.101256
39. Bharadwaj S, Ballare S, Rohit, Chandel MK. Impact of congestion on greenhouse gas emissions for road transport in Mumbai metropolitan region. *Transportation Research Procedia*. 2017 Jan 1;World Conference on Transport Research - WCTR 2016 Shanghai. 10-15 July 201625:3538–51. doi:10.1016/j.trpro.2017.05.282
40. Li J, Huang Y, Zhang C, Yao D. How does public transport development contribute to carbon emission reduction? *Transportation Research Part A: Policy and Practice*. 2025 Jan 1;191:104327. doi:10.1016/j.tra.2024.104327
41. Lu IJ, Lin SJ, Lewis C. Decomposition and decoupling effects of carbon dioxide emission from highway transportation in Taiwan, Germany, Japan and South Korea. *Energy Policy*. 2007 Jun 1;35(6):3226–35. doi:10.1016/j.enpol.2006.11.003
42. Hansen WG. How Accessibility Shapes Land Use. *Journal of the American Institute of Planners*. 1959 May 1;25(2):73–6. doi:10.1080/01944365908978307
43. Soukhov A, Pereira RHM, Higgins CD, Páez A. A family of accessibility measures derived from spatial interaction principles. *PLOS ONE*. 2025 Nov 14;20(11):e0335951. doi:10.1371/journal.pone.0335951
44. El-Geneidy A, Levinson D. Making accessibility work in practice. *Transport Reviews*. 2022 Mar 4;42(2):129–33. doi:10.1080/01441647.2021.1975954

45. Palm M, Teel S, Tiznado-Aitken I, Soukhov A, Paez A, Farber S, et al. Developing Data Driven Equity Standards: Stakeholder Perspectives [Internet]. Mobilizing Justice; 2023 May [cited 2023 Dec 12]. p. 26. Report no.: 1. Available from: https://mobilizingjustice.ca/wp-content/uploads/2023/06/Developing_Data_Driven_Standards_FINAL_2023_06_06.pdf
46. Martens K. Transportation Planning Based on Principles of Justice. In: Transport Justice. Routledge; 2016.
47. Wu H, Avner P, Boisjoly G, Braga CKV, El-Geneidy A, Huang J, et al. Urban access across the globe: an international comparison of different transport modes. *npj Urban Sustain*. 2021 May 27;1(1):16. doi:10.1038/s42949-021-00020-2
48. Ryan J, Martens K. Defining and implementing a sufficient level of accessibility: What's stopping us? *Transportation Research Part A: Policy and Practice*. 2023 Sep 1;175:103792. doi:10.1016/j.tra.2023.103792
49. Wee B van, Geurs K. Discussing Equity and Social Exclusion in Accessibility Evaluations. *European Journal of Transport and Infrastructure Research*. 2011 Sep 1;11(4):4. doi:10.18757/ejtir.2011.11.4.2940
50. Marc Steen, Menno Manschot, Nicole De Koning. Benefits of Co-design in Service Design Projects. *International Journal of Design* [Internet]. 2011 [cited 2026 Sep 8];5(2). Available from: <https://www.ijdesign.org/index.php/IJDesign/article/view/890/346>
51. Sánchez De La Guía L, Puyuelo Cazorla M, de-Miguel-Molina B. Terms and meanings of “participation” in product design: From “user involvement” to “co-design.” *The Design Journal*. 2017 Jul 28;20(sup1):S4539–51. doi:10.1080/14606925.2017.1352951
52. Vargas C, Zorbas C, Longworth GR, Ugalde A, Needham C, Sunil A, et al. Exploring co-design: a systematic review of concepts, processes, models, and frameworks used in public health research. *Journal of Public Health*. 2025 Dec 1;47(4):e616–39. doi:10.1093/pubmed/fdaf084
53. DeLuca PF, Kanaroglou PS, DeLuca PF, Kanaroglou PS. Code Red: Explaining Average Age of Death in the City of Hamilton. *AIMSPH*. 2015;2(4):730–45. doi:10.3934/publichealth.2015.4.730
54. Hasso Plattner Institute of Design. An Introduction to Design Thinking PROCESS GUIDE [Internet]. Hasso Plattner Institute of Design. Available from: <https://web.stanford.edu/~mshanks/MichaelShanks/files/509554.pdf>
55. McMaster University. Equity-based Co-Creation Hub [Internet]. 2026 [cited 2026 Sep 8]. McMaster University's Equity-based Co-Creation Hub. Available from: <https://codesign.mcmaster.ca/>
56. Jens von B, Dmitry S, Jacobs A. cancensus: R package to access, retrieve, and work with Canadian Census data and geography [Internet]. 2026. Available from: <https://mountainmath.github.io/cancensus/>
57. Pereira RHM, Saraiva M, Herszenhut D, Braga CKV, Conway MW. r5r: Rapid Realistic Routing on Multimodal Transport Networks with R⁵ in R. *Findings*. 2021 Mar 4. doi:10.32866/001c.21262
58. Bhagat-Conway M, Palm M. {gtfsrealtime}: a Package for Working with GTFS-Realtime Transit Data in R. *Findings*. 2026 Aug 31. doi:10.32866/001c.168472
59. Soukhov A, Páez A. TTS2016R: A data set to study population and employment patterns from the 2016 Transportation Tomorrow Survey in the Greater Golden Horseshoe area, Ontario, Canada. *Environment and Planning B: Urban Analytics and City Science*. 2023 Feb;50(2):556–63. doi:10.1177/23998083221146781

60. Paez A, Robin. paezha/spatial-analysis-R: Pre-release to obtain doi from Zenodo [Internet]. Zenodo; 2021 [cited 2026 Sep 12]. Available from: <https://zenodo.org/record/5155983> doi:10.5281/ZENODO.5155983
61. Bertram MG, Sundin J, Roche DG, Sánchez-Tójar A, Thoré ESJ, Brodin T. Open science. *Current Biology*. 2023 Aug;33(15):R792–7. doi:10.1016/j.cub.2023.05.036
62. Arribas-Bel D, Green M, Rowe F, Singleton A. Open data products-A framework for creating valuable analysis ready data. *J Geogr Syst*. 2021 Oct;23(4):497–514. doi:10.1007/s10109-021-00363-5
63. Santos BDD, Moghadasi M, Páez A. ActiveCA: Time use data from the general social survey of Canada to study active travel. *Environment and Planning B: Urban Analytics and City Science*. 2025 Oct;52(8):2037–47. doi:10.1177/23998083251374724
64. Desjardins E, Higgins CD, Páez A. Examining equity in accessibility to bike share: A balanced floating catchment area approach. *Transportation Research Part D: Transport and Environment*. 2022 Jan;102:103091. doi:10.1016/j.trd.2021.103091
65. Páez A. Open spatial sciences: an introduction. *J Geogr Syst*. 2021 Oct;23(4):467–76. doi:10.1007/s10109-021-00364-4
66. Karner A, Pereira RHM, Farber S. Advances and pitfalls in measuring transportation equity. *Transportation (Dordrecht)*. 2025;52(4):1399–427. doi:10.1007/s11116-023-10460-7
67. Herszenhut D, Pereira RHM, Portugal L da S, Oliveira MH de S. The impact of transit monetary costs on transport inequality. *Journal of Transport Geography*. 2022 Feb 1;99:103309. doi:10.1016/j.jtrangeo.2022.103309
68. Fink C, Klumpenhower W, Saraiva M, Pereira R, Tenkanen H. r5py: Rapid Realistic Routing with R5 in Python [Internet]. Zenodo; 2022 [cited 2026 Sep 12]. Available from: <https://zenodo.org/record/7060438> doi:10.5281/ZENODO.7060438
69. Higgins CD, Li C, Páez A. Person-based aggregate space-time accessibility (PASTA): Bridging the gap between place- and person-based accessibility. *Journal of Transport Geography*. 2026 Apr 1;132:104602. doi:10.1016/j.jtrangeo.2026.104602
70. Soukhov A, Páez A, Higgins CD, Mohamed M. Introducing spatial availability, a singly-constrained measure of competitive accessibility. *Plos One*. 2023;18(1):e0278468. doi:10.1371/journal.pone.0278468
71. Parga JPFA, Soukhov A, Arku RN, Higgins CD, Páez A. Democratic Access to our Cities: The impacts of recent changes to transit services in major Canadian metropolitan areas [Internet]. 2025 Jan [cited 2026 Sep 23]. Available from: <https://hdl.handle.net/1807/152158>
72. Higgins CD, Páez A, Kim G, Wang J. Changes in accessibility to emergency and community food services during COVID-19 and implications for low income populations in Hamilton, Ontario. *Social Science & Medicine*. 2021 Dec 1;291:114442. doi:<https://doi.org/10.1016/j.socscimed.2021.114442>
73. Arku RN, Higgins CD, Farber S. Untangling proximity and accessibility effects of transit on property prices. *Transport Policy*. 2025 Dec 1;174:103852. doi:10.1016/j.tranpol.2025.103852
74. Higgins CD. pycnogrid [R] [Internet]. 2026 [cited 2026 Sep 12]. Available from: <https://github.com/higgicd/pycnogrid>

75. De Barros J. Reproducing the British Caribbean: Sex, Gender, and Population Politics after Slavery [Internet]. Chapel Hill, UNITED STATES: University of North Carolina Press; 2014 [cited 2026 Sep 20]. Available from: <http://ebookcentral.proquest.com/lib/mcmu/detail.action?docID=1693182>
76. Brown JR, Morris EA, Taylor BD. Planning for Cars in Cities: Planners, Engineers, and Freeways in the 20th Century. *Journal of the American Planning Association*. 2009;75(2):161–77. doi:10.1080/01944360802640016
77. Geva Y, Lambropoulos A, Mendita I, Siemiatycki M. Missed Connections: Locating equity in Toronto's transportation history [Internet]. Infrastructure Institute, University of Toronto; 2025. Available from: <https://infrastructureinstitute.ca/missed-connections/>
78. Mahmoud M, Hine J, Kashyap A. Stakeholders' perspectives towards bus quality indicators. *Proceedings of the Institution of Civil Engineers - Transport*. 2014 Jun;167(3):156–77. doi:10.1680/tran.11.00083
79. Tasnim S, Mohamed M. Fixed-route or demand-responsive transit? An evaluation framework for transit service structures using dual-perspective indicators. *Journal of Public Transportation*. 2026;28:100161. doi:10.1016/j.jpuptr.2026.100161
80. Eldeeb G, Mohamed M. Consumers oriented investments in transit service quality improvements: The best bang for your buck. *Research in Transportation Economics*. 2022 Sep;94:101175. doi:10.1016/j.retrec.2021.101175
81. Eldeeb G, Sears S, Mohamed M. What do users want from transit? Qualitative analysis of current and potential users' perceptions. *Transportation Research Part A: Policy and Practice*. 2023 May;171:103649. doi:10.1016/j.ra.2023.103649
82. Sears S, Moataz M, Ferguson M, Razavi S, Páez A. Perceived barriers to the movement of goods in Canada: A grounded theory investigation. *Transportation Research Part A: Policy and Practice*. 2022 Aug 1;162:27–45. doi:<https://doi.org/10.1016/j.ra.2022.05.011>
83. Ravensbergen L, Varickanickal J, Newbold KB, Mohamed M. A mixed-methods investigation of older adults' public transit use and travel satisfaction. *Travel Behaviour and Society*. 2023 Jul 1;32:100577. doi:10.1016/j.tbs.2023.100577
84. Eldeeb G, Mohamed M. Understanding the Transit Market: A Persona-Based Approach for Preferences Quantification. *Sustainability*. 2020 May 8;12(9). doi:10.3390/su12093863
85. Eldeeb G, Mohamed M. How do psychological factors sway customers' preferences towards transit service attributes? *Case Studies on Transport Policy*. 2022 Jun 1;10(2):905–14. doi:10.1016/j.cstp.2022.03.003
86. Eldeeb G, Mohamed M. Quantifying preference heterogeneity in transit service desired quality using a latent class choice model. *Transportation Research Part A: Policy and Practice*. 2020 Sep 1;139:119–33. doi:10.1016/j.ra.2020.07.006
87. Eldeeb G, Mohamed M, Páez A. Built for active travel? Investigating the contextual effects of the built environment on transportation mode choice. *Journal of Transport Geography*. 2021 Oct;96:103158. doi:10.1016/j.jtrangeo.2021.103158
88. Higgins CD, Adams MD, Réquia W, Mohamed M. Accessibility, Air Pollution, and Net Land Value Uplift from Highway Infrastructure. Vol. 84. 2018;84:177–91.
89. Shehabeldeen A, Mohamed M. Factors Influencing Electric Vehicle Adoption: Coupling Machine Learning Models and Open-Source Data - Ali Shehabeldeen, Moataz Mohamed, 2026. Vol. 2680. 2026;2680(2):948–66. doi:<https://doi.org/10.1177/03611981251368317>

90. Moniruzzaman M, Paez A. An investigation of the attributes of walkable environments from the perspective of seniors in Montreal. *Journal of Transport Geography*. 2016 Feb;51:85–96. doi:10.1016/j.jtrangeo.2015.12.001
91. Moniruzzaman M, Paez A, Scott D, Morency C. Trip generation of seniors and the geography of walking in Montreal. *Environment and Planning A*. 2015;47(4):957–76. doi:10.1068/a130070p
92. Páez A, Mercado R, Farber S, Morency C, Roorda M. Accessibility to health care facilities in Montreal Island: An application of relative accessibility indicators from the perspective of senior and non-senior residents. *International Journal of Health Geographics*. 2010;9(52):1–9.
93. Paez A, Scott D, Potoglou D, Kanaroglou P, Newbold KB. Elderly mobility: Demographic and spatial analysis of trip making in the Hamilton CMA, Canada. *Urban Studies*. 2007 Jan;44(1):123–46. doi:10.1080/00420980601023885
94. Roorda MJ, Paez A, Morency C, Mercado R, Farber S. Trip generation of vulnerable populations in three Canadian cities: a spatial ordered probit approach. *Transportation*. 2010 May;37(3):525–48. doi:10.1007/s11116-010-9263-3
95. Páez A, Farber S, Mercado R, Roorda M, Morency C. Jobs and the Single Parent: An Analysis of Accessibility to Employment in Toronto. *Urban Geography*. 2013 Sep 1;34(6):815–42. doi:10.1080/02723638.2013.778600
96. Reyes M, Paez A, Morency C. Walking accessibility to urban parks by children: A case study of Montreal. *Landscape and Urban Planning*. 2014 May;125:38–47. doi:10.1016/j.landurbplan.2014.02.002
97. Tavakoli Z, Abdollahi S, Waygood EOD, Páez A, Boisjoly G. Traffic danger's potential impact on children's accessibility. *Transportation Research Part D: Transport and Environment*. 2024 Oct 1;135:104370. doi:10.1016/j.trd.2024.104370
98. Tavakoli Z, Waygood O, Abdollahi S, Paez A. “Where Do Children Go?”: Exploring Children's Daily Destinations With Children, Parents, and Experts. *Urban Planning*. 2024 Oct 31;9(0). doi:10.17645/up.8478
99. Páez A, Scott DM, Morency C. Measuring accessibility: positive and normative implementations of various accessibility indicators. *Journal of Transport Geography*. 2012 Nov 1;Special Section on Accessibility and Socio-Economic Activities: Methodological and Empirical Aspects25:141–53. doi:10.1016/j.jtrangeo.2012.03.016
100. Páez A, Gertes Mercado R, Farber S, Morency C, Roorda M. Relative Accessibility Deprivation Indicators for Urban Settings: Definitions and Application to Food Deserts in Montreal. *Urban Studies*. 2010 Jun 1;47(7):1415–38. doi:10.1177/0042098009353626
101. Farber S, Bartholomew K, Li X, Paez A, Habib KMN. Assessing social equity in distance based transit fares using a model of travel behavior. *Transportation Research Part A-Policy and Practice*. 2014 Sep;67:291–303. doi:10.1016/j.tra.2014.07.013
102. Soukhov A, Tarrío-Ortiz J, Soria-Lara JA, Páez A. Multimodal spatial availability: A singly-constrained measure of accessibility considering multiple modes. *PLOS ONE*. 2024;19(2):e0299077. doi:10.1371/journal.pone.0299077
103. Dos Santos BD, Trevor Hanson, Antonio Páez. *Dignity in Transportation: A Scoping Review and Theoretical Framework*. 2026.

104. Páez A. Exploring contextual variations in land use and transport analysis using a probit model with geographical weights. *Journal of Transport Geography*. 2006 May 1;14(3):167–76. doi:10.1016/j.jtrangeo.2005.11.002
105. Páez A, Moniruzzaman Md, Bourbonnais PL, Morency C. Developing a web-based accessibility calculator prototype for the Greater Montreal Area. *Transportation Research Part A: Policy and Practice*. 2013 Dec 1;58:103–15. doi:10.1016/j.tra.2013.10.020
106. Moniruzzaman M, Chudyk A, Páez A, Winters M, Sims-Gould J, McKay H. Travel behavior of low income older adults and implementation of an accessibility calculator. *Journal of Transport & Health*. 2015 Jun 1;2(2):257–68. doi:10.1016/j.jth.2015.02.006
107. Desjardins E, Apatu E, Razavi SD, Higgins CD, Scott DM, Páez A. “Going through a little bit of growing pains”: A qualitative study of the factors that influence the route choice of regular bicyclists in a developing cycling city. *Transportation Research Part F: Traffic Psychology and Behaviour*. 2021 Aug 1;81:431–44. doi:https://doi.org/10.1016/j.trf.2021.06.005
108. Ravensbergen L, Newbold KB, Ganann R. ‘It’s overwhelming at the start’: transitioning to public transit use as an older adult. *Ageing & Society*. 2024 Jan;44(1):43–60. doi:10.1017/S0144686X22000010
109. Ravensbergen L, Newbold KB, Ganann R, Sinding C. ‘Mobility work’: Older adults’ experiences using public transportation. *Journal of Transport Geography*. 2021 Dec 1;97:103221. doi:10.1016/j.jtrangeo.2021.103221
110. Ravensbergen L, Fournier J, El-Geneidy A. Exploratory Analysis of Mobility of Care in Montreal, Canada. *Transportation Research Record: Journal of the Transportation Research Board*. 2023 Jan;2677(1):1499–509. doi:10.1177/03611981221105070
111. Ravensbergen L, Schwanen T. Community Transport’s Dual Role as a Transport and a Social Scheme: Implications for Policy. *International Journal of Environmental Research and Public Health*. 2024 Mar 29;21(4). doi:10.3390/ijerph21040422
112. Ravensbergen L, Schwanen T. Community Transport’s Dual Role as a Transport and a Social Scheme: Implications for Policy. *IJERPH*. 2024 Mar 30;21(4):422. doi:10.3390/ijerph21040422
113. Ravensbergen L, Schwanen T. “*It’s such a big ask for a volunteer*”: Community transport as a sticking plaster solution to rural transport poverty. *Journal of Rural Studies*. 2025 Oct 1;119:103718. doi:10.1016/j.jrurstud.2025.103718
114. Full article: ‘I wouldn’t take the risk of the attention, you know? Just a lone girl biking’: examining the gendered and classed embodied experiences of cycling [Internet]. [cited 2026 Sep 21]. Available from: <https://www.tandfonline.com/doi/full/10.1080/14649365.2020.1806344>
115. Ravensbergen L, Ilunga-Kapinga J, Ismail S, Patel A, Khachatryan A, Wong K. Cycling as social practice: a collective autoethnography on power and vélomobility in the city. *Mobilities*. 2024 Mar 3;19(2):329–43. doi:10.1080/17450101.2023.2211238
116. Ravensbergen L, Schwanen T. Care-driven informality: The case of community transport. *Geographical Journal*. 2024 Jun;190(2):e12552. doi:10.1111/geoj.12552
117. Theodore Collins, Karen Van Ooteghem, Sheereen Harris, Marla Beauchamp, Léa Ravensbergen. “I’d Like to See All of It”: Co-Designing a Feedback Mobility Report with Older Adults. *Journal of Population Ageing*. Under review.

118. Alexandra Sbrocchi, Liam Bienstock, Elyse Comeau, Jacob Alhassan, Marvin Macaraig, Lea Ravensbergen, et al. Epistemic Injustice in Community-Based Transportation Research: Insights from Researchers' Experiences. *Journal of Transport Geography*. Under review.
119. Léa Ravensbergen, Darren Scott, Bruce Newbold. Mitacs [Internet]. [cited 2026 Sep 21]. Accessible Transportation Services: Customer Travel Satisfaction. Available from: <https://www.mitacs.ca/our-projects/accessible-transportation-services-customer-travel-satisfaction/>
120. Collins T, Ravensbergen L, Young M. The landscape of contemporary paratransit research: A critical systematic review of the literature in the US and Canada. *Travel Behaviour and Society*. 2025 Jan 1;38:100930. doi:10.1016/j.tbs.2024.100930
121. Paez A. Reproducible Research Workflow [HTML] [Internet]. 2026 [cited 2026 Sep 12]. (Notes and resources for graduate course GEOG 712 Reproducible Research Workflow with GitHub and R). Available from: <https://github.com/paezha/Reproducible-Research-Workflow>